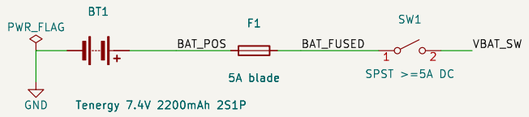
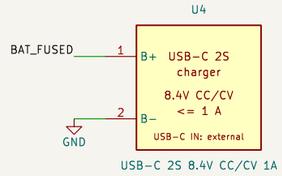
RoomCleaner claw — Rev B

Point-to-point wired claw. Net names and connections = the net table of CLAW_REVB_SPEC.md (drawn from claw_spec.py). Design, not yet built or bench-tested.

BATTERY, FUSE, MAIN SWITCH (load side) and CHARGER (pack side)

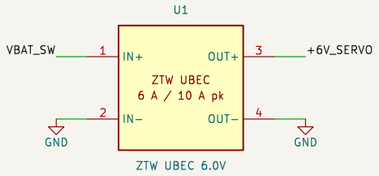




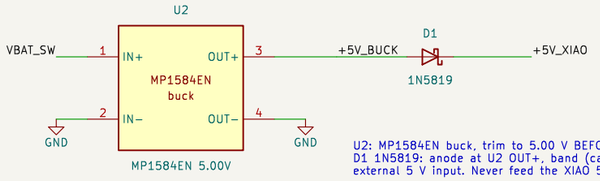
U4: USB-C input from a 5 V adapter is external (not drawn). CHARGE ONLY WITH SW1 OFF. Charger must be an 8.4 V CC/CV type at <= 1 A (Tenergy limit).

BT1: Tenergy 7.4 V 2200 mAh 2S1P Li-ion with protection PCB (31003 assumed: 3 A continuous, 1 A max charge); 8.4 V full, 6.8 V firmware cutoff. Confirm label and connector.
F1: 5 A blade fuse in a holder on the battery + lead (provisional); protects the wiring, the pack PCB is the cell protection. SW1: SPST >= 5 A DC, load side only; the charger stays on the pack side.

CONVERTERS: servo rail (UBEC 6.0 V) and logic rail (MP1584 5.00 V through D1)

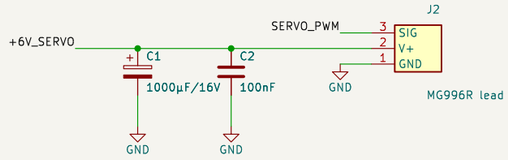


U1: ZTW UBEC (6 A continuous / 10 A peak), jumper set to 6.0 V. Servo only. Measure the output before connecting the servo.



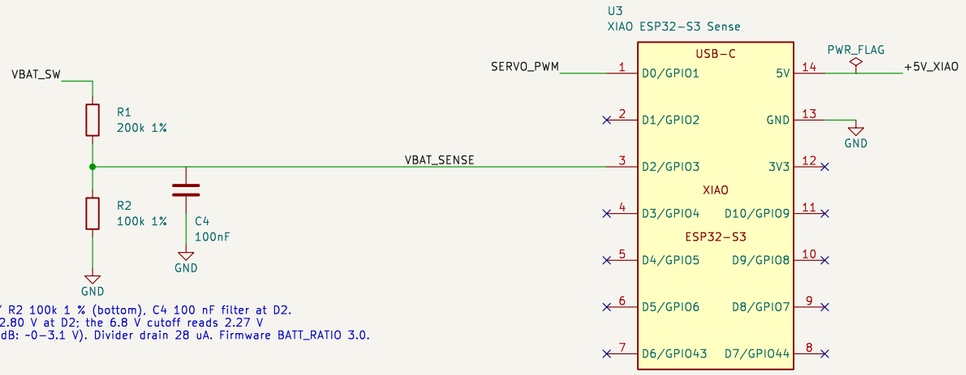
U2: MP1584EN buck, trim to 5.00 V BEFORE connecting the XIAO. XIAO only.
D1 1N5819: anode at U2 OUT+, band (cathode) toward the XIAO; Seeed requires a diode on external 5 V input. Never feed the XIAO 5V pad from the 6 V rail.

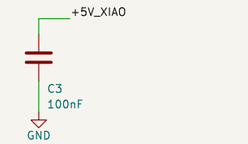
SERVO CONNECTOR AND DECOUPLING



C1 1000 uF / 16 V (+ to V+) and C2 100 nF sit at the servo connector.
J2 (MG996R lead): pin 1 GND (brown), pin 2 V+ (red), pin 3 signal (orange); check your lead.
Signal is 3.3 V PWM from D0 / GPIO1 into a 6 V servo (no buffer, as the firmware assumes).

LOGIC: XIAO ESP32-S3 Sense, battery sense divider (D2 / GPIO3), 5 V decoupling





C3 100 nF at the XIAO 5V pad.
PWR_FLAG: 5 V reaches the pad through D1 (or from USB-C when programming).

R1 200k 1 % (top) / R2 100k 1 % (bottom), C4 100 nF filter at D2.
8.4 V x 100/300 = 2.80 V at D2; the 6.8 V cutoff reads 2.27 V
(ESP32-S3 ADC, 11 dB: ~0-3.1 V). Divider drain 28 uA. Firmware BATT_RATIO 3.0.

UNCONNECTED (per spec): XIAO D1, D3-D10, 3V3 (no-connect flags), BAT pads;
U1 / U2 have no enable pins on these symbols; U4 USB-C input is the external adapter.

SENSE CAMERA BOARD (note only, not wired here)
The Sense camera board is attached to the XIAO ESP32-S3 by its factory B2B connector; no header pin is used for it. BAT pads on the underside of the XIAO are unused.
USB-C on the XIAO is for programming (5V pad is fed from D1; never from the 6 V rail).

D31 AND AUTHORITY

DECISIONS.md D31 (2026-10-21): Tenergy 7.4 V 2S pack; ZTW UBEC at 6.0 V for the servo; MP1584 at 5.00 V for the XIAO through a 1N5819, never from the 6 V rail; USB-C 2S charger on the pack side of the main switch.
Spec: docs/electrical/claw_revB_20261021/CLAW_REVB_SPEC.md. Pin map: firmware/claw_s3/claw_s3.ino (servo PWM D0 / GPIO1; battery sense D2 / GPIO3, BATT_RATIO 3.0, cutoff 6.8 V).

CHARGING RULE (D31)

CHARGE ONLY WITH SW1 OFF, attended, on a non-flammable surface, with an 8.4 V CC/CV charger at <= 1 A.
The pack's protection PCB must provide per-cell overcharge / over-discharge protection: confirm it on the label or datasheet before the first charge (a 2-wire pack cannot be balance-charged).

EMERGENCY STOP

The emergency stop is the switched power strip at the stations. SW1 on the claw is a SERVICE DISCONNECT only; it does not stop the winches.

CHECKS (calculated, spec)

- Battery sense: 8.4 V x 100/300 = 2.80 V at D2; 6.8 V cutoff -> 2.27 V. Inside the ESP32-S3 ADC range at 11 dB (~0-3.1 V). Divider drain 28 uA.
- Peak pack current: servo stall ~2.5 A at 6 V (15 W, provisional) + logic ~0.7 A at 5 V (3.5 W) = 18.5 W; at 88 % efficiency 2.8 A at 7.4 V, 3.1 A at 6.8 V: at the pack's ~3 A continuous rating. Stalls are short, but the pack PCB may trip on a long stall near empty. Measure the servo stall current.
- Servo signal is 3.3 V PWM into a 6 V servo (no buffer, as the firmware assumes). Most MG996R units accept it; bench-check the full open/close travel.
- UBEC near empty: 6.8 V in for 6.0 V out leaves little headroom: the servo rail may sag near cutoff, acceptable because the firmware stops lifts at 6.8 V.

BEFORE THE FIRST POWER-UP

1. Set U1 to 6.0 V and U2 to 5.00 V with a meter, nothing connected to their outputs.
2. Confirm U4 is an 8.4 V CC/CV module and its charge current is <= 1 A (Tenergy limit).
3. Check the pack voltage (7.4-8.4 V) and polarity at the connector.
4. Power up with the servo unplugged; check D2 reads ~VBAT/3 in /status.
5. Plug in the servo; run /setup to park it open.

Design, not yet built or bench-tested
Net table of CLAW_REVB_SPEC.md; DECISIONS.md D31
RoomCleaner

Sheet: /
File: RoomCleaner_Claw_RevB.kicad_sch

Title: RoomCleaner claw — Rev B

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1